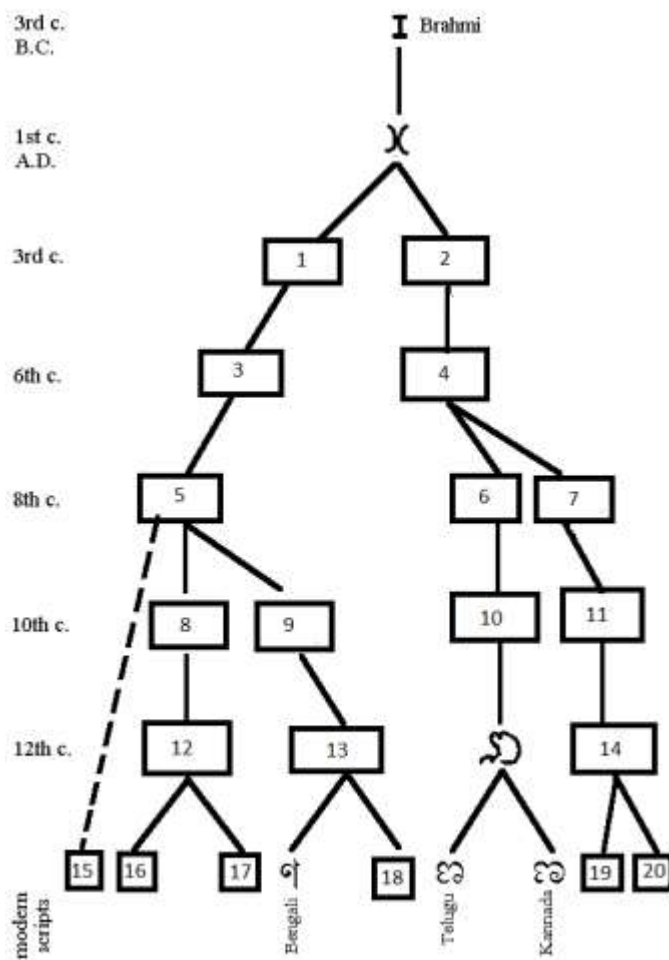


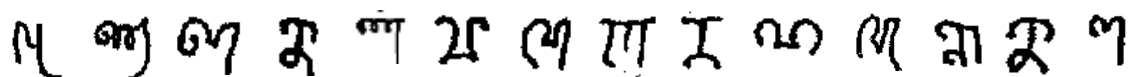
Problem #1: Evolution of *Nna* (15 marks)



The phylogenetic tree depicting the evolution of the glyphs for the grapheme *Nna* (retroflex nasal η)

Assignment 1

Your task is to complete the above phylogenetic tree by filling the boxes with the appropriate glyphs given below or the modern forms written above.



A B C D E F G H I J K L M N

Write down your answer for each box by specifying the letter (A-N) of the glyph that goes in the box. For modern forms, use the language/script name.

All the modern Indian language scripts have evolved from an ancient script called *Brahmi*, which was in use around 3rd century BC. Observe the forms (*glyphs*) of the letter (*grapheme*) *Nna* in 9 modern

Indian scripts: Devanagari: **५**, Bengali:

જ, Gujarati: જ્ઞ, Gurumukhi (Punjabi):

𑌕, Kannada: ಐ, Malayalam: എ

Odiya: **ଈ** , Tamil: **ஐ** Telugu: **ఐ**.

Linguists believe that all these forms have developed from the Brahmi glyph **I**. Based on the several intermediate glyphs of this grapheme that has been found in the historical documents, linguists have been able to create the evolutionary history of the grapheme *Nna*, represented in the form of a *phylogenetic tree* or *evolutionary tree* as shown in the left. In this tree, bottom-most boxes denote the glyphs in the modern scripts and higher boxes denote older forms from which the later glyphs have evolved. (Most of the intermediate glyphs have been omitted and replaced by numbers.) For instance, glyph #9 (used in 10th century) has given rise to glyph #13 (used in 12th century) which in turn has led to the development of the glyphs used in modern Bengali and

Assignment 2

Modern scripts and the glyphs have remained almost fixed for over 300 years now, and are unlikely to change in the future. Invention of which revolutionary technology in the 15th century has led to this situation? Explain how this technology has prevented in further evolution of the structure of letters.

Brāhmī is one of the oldest writing systems used in the Indian subcontinent and in Central Asia during the final centuries BCE and the early centuries CE. It is the ancestor of all the modern Indic scripts as well as the Thai, Lao, Tibetan and Burmese scripts. In modern times, it was deciphered by James Prinsep, a British archaeologist, in 1837.

Nna is the Unicode name for the letter representing the *nasal retroflex* /ɳ/ – a nasal sound produced by curling the tongue. Retroflex sounds, such as /ɳ/, /ʈ/ and /ɖ/ are found in several Indic and South Asian languages.

- *Monojit Choudhury*